Lecture 1

Introduction to Computer Systems and Number Systems

GOLDEN GATE UNIVERSITY

Computer Organization and Digital Foundations

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Today's Roadmap

1. Describe the main components and layers of a computer system
2. Explain why digital computers use binary
3. Read and write numbers in binary, octal, decimal and hexadecimal
4. Convert between number bases by hand

60-minute plan

0-5 min	Orientation
5-25 min	Computer system overview
25-50 min	Number systems and conversions
50-60 min	Practice, real life, wrap-up

What Is a Computer System?

HW

Hardware

Physical parts: processor (CPU), memory, storage, input and output devices.

SW

Software

Programs: operating system, compilers, applications and device drivers.

D

Data

Information as bit patterns: numbers, text, images, sound, video.

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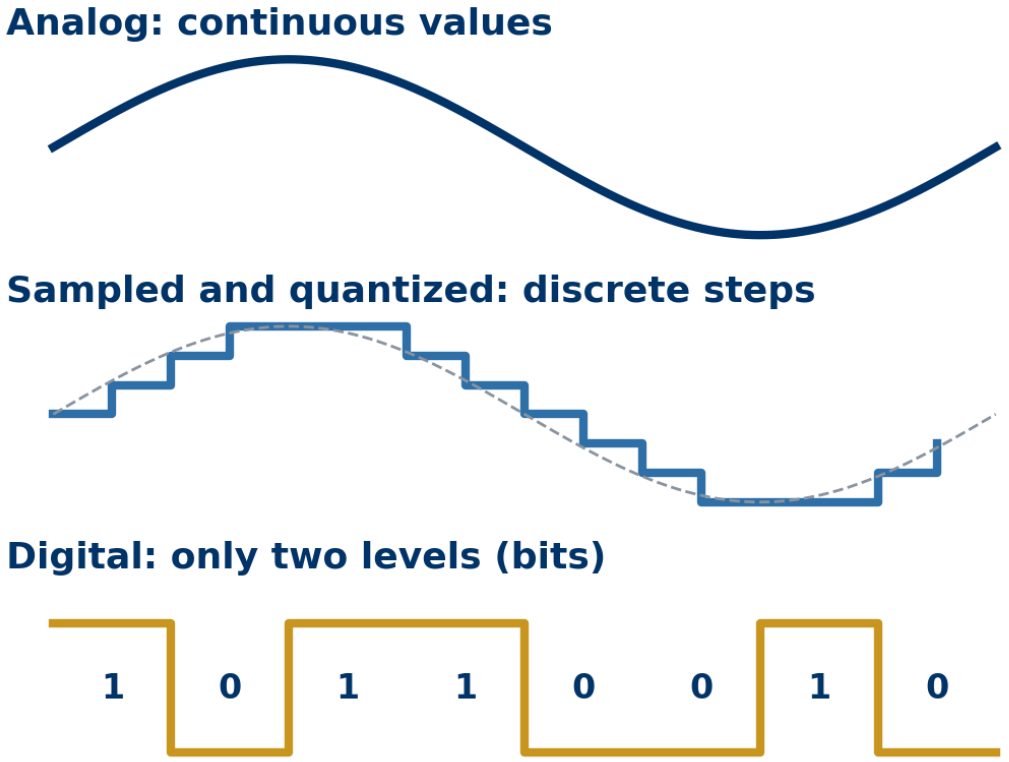
Users and environment

People or other systems that supply input and use the output.

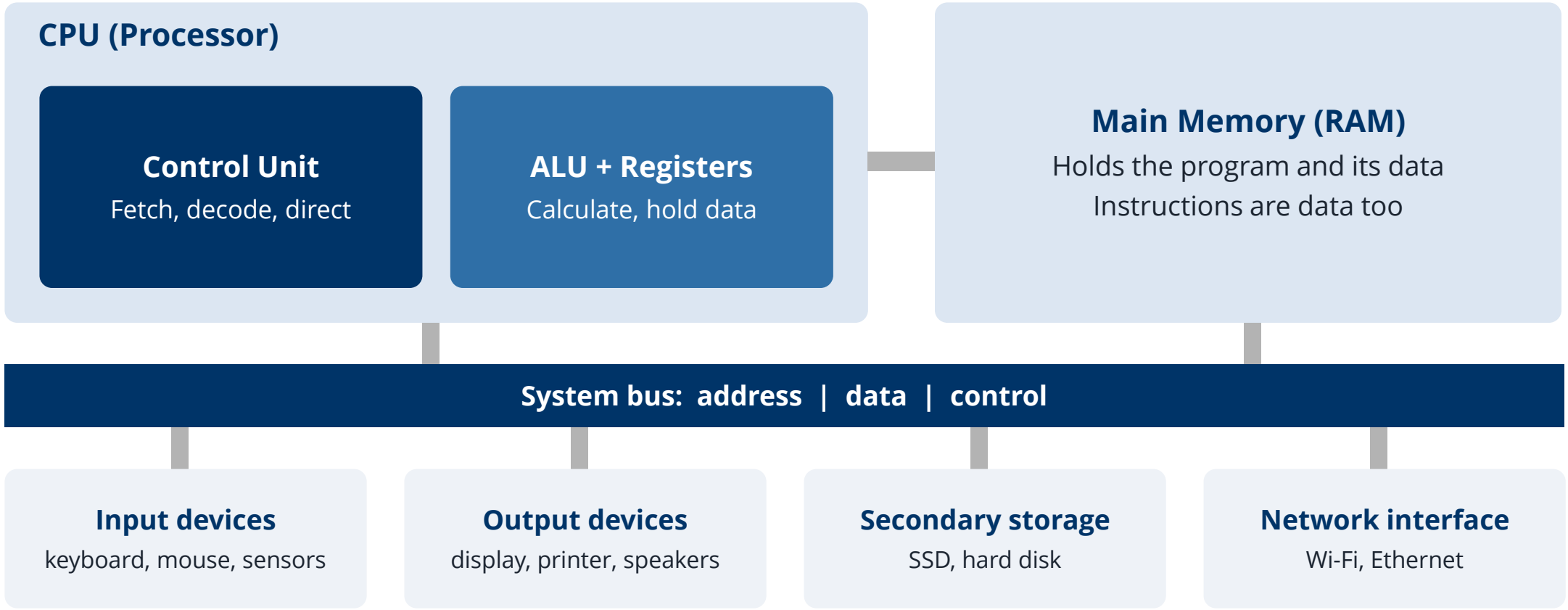
A computer system is hardware and software cooperating to process data.

Digital versus Analog

- **Analog:** signals vary continuously, e.g. a thermometer or microphone voltage
- **Digital:** information is a sequence of discrete symbols; in hardware, just two values
- **Conversion:** sample, quantise, then encode as bits
- **Why digital wins:** noise immunity, exact copies, one machine runs any program



Block Diagram of a Computer



Layers of Abstraction

Application software	Word processors, browsers, games
Language and compiler	C, Java, Python translated to machine code
Operating system	Manages processes, memory, files, devices
Instruction set architecture	Hardware-software interface: x86-64, ARM, RISC-V
Microarchitecture	Pipelines, caches, buses, number of cores
Digital logic	**Gates, flip-flops, binary data (our focus now)**
Transistors and circuits	Semiconductor physics, voltages, timing

Software layers (blue) | Interface (gold) | Hardware layers (navy)

Classes of Computers

Class	Typical examples	Main design priorities
Personal mobile device	Smartphones, tablets	Cost, energy efficiency, responsive media
Desktop	Desktop PCs, laptops, netbooks	Price-performance, energy, graphics
Server	Web, database, file servers	Availability, scalability, throughput
Cluster / warehouse-scale	Cloud services, web search	Price-performance, power, throughput
Embedded	Cars, appliances, routers	Price, power, application-specific speed

1.8 bn
PMDs sold in 2010

350 m
desktop PCs

20 m
servers

19 bn
embedded processors

What Is Computer Architecture?

Instruction Set Architecture

The programmer-visible interface: instructions, registers, addressing modes, data types.

x86-64, ARM, MIPS, RISC-V

Organization (microarchitecture)

How the CPU and memory system are built to implement the ISA.

Pipelines, caches, buses, cores

Hardware

Detailed logic design, circuit technology and packaging of a specific chip.

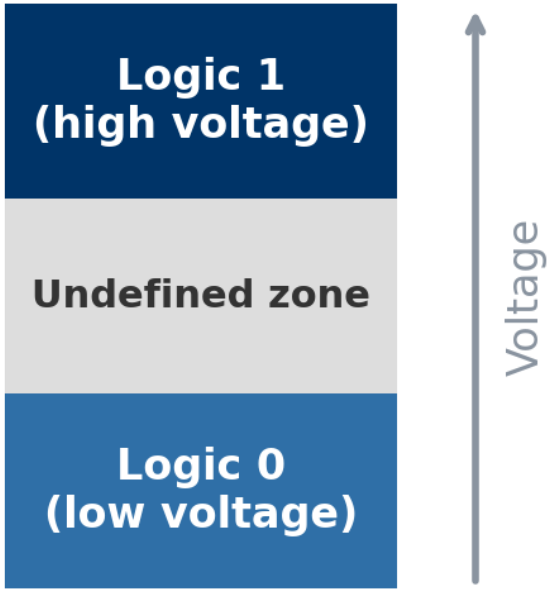
Transistors, clock rate, power, layout

Same ISA, different implementations

Two processors from different companies can both implement x86-64. Programs run unchanged on both, but speed, power and cost differ because organisation and hardware differ.

Why Computers Use Binary

- Transistors switch reliably between two states: ON and OFF
- Two wide voltage ranges give a noise margin, so small disturbances do not flip a bit
- Two-valued logic (true/false) matches Boolean algebra and logic gates
- Numbers, text, images and sound can all be encoded as bit patterns



1 bit
a single 0 or 1

1 byte
8 bits = 256 patterns

1 word
32 or 64 bits on modern CPUs

Positional Number Systems

Value of a number = Σ ($d_i \times r^i$) digit d at position i, radix r

Decimal
radix r = 10

10^3	10^2	10^1	10^0
5	**2**	**7**	**4**

5×1000 + 2×100 + 7×10 + 4×1
= 5,274

Binary
radix r = 2

2^3	2^2	2^1	2^0
1	**0**	**1**	**1**

1×8 + 0×4 + 1×2 + 1×1
= 11_{10}

Same rule in every base: only the radix r changes.

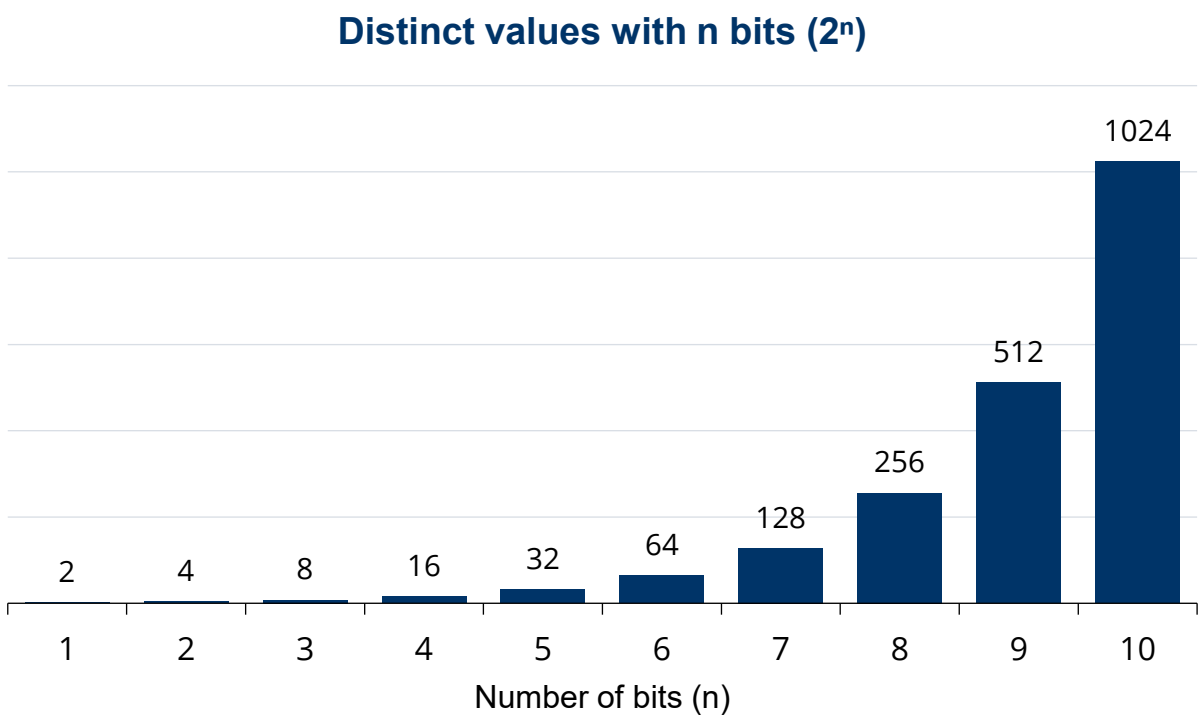
Four Number Systems at a Glance

Dec	Binary	Octal	Hex
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7

Dec	Binary	Octal	Hex
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Decimal (base 10) **Binary (base 2)** **Octal (base 8)** **Hexadecimal (base 16)**

Bits, Bytes and Powers of Two



8 bits
256 values (0-255)

16 bits
65,536 values

32 bits
about 4.29 billion

64 bits
about 1.8×10^{19}

Rule of thumb

n bits give 2^n patterns; largest unsigned value is $2^n - 1$

Memory tip: 1 KB = 1,024 bytes = 2^{10} bytes (strictly one kibibyte, KiB).

Binary to Decimal Conversion

Example 1: 101101_2

Weight	32	16	8	4	2	1
Bit	1	0	1	1	0	1
Value	32	0	8	4	0	1

$32 + 8 + 4 + 1 = 45_{10}$

Example 2: 1101.101_2

Weight	8	4	2	1		½	¼	⅛
Bit	1	1	0	1	.	1	0	1
Value	8	4	0	1		0.5	0	0.125

$13 + 0.625 = 13.625_{10}$

1. Write the weight (2^i) above every bit
2. Keep the weight where the bit is 1
3. Add all the kept weights

Decimal to Binary Conversion

Integer part: 45_{10} (divide by 2)

Division	Quotient	Remainder
45 ÷ 2	22	1
22 ÷ 2	11	0
11 ÷ 2	5	1
5 ÷ 2	2	1
2 ÷ 2	1	0
1 ÷ 2	0	1

Read remainders bottom → top: 101101_2



Fraction part: 0.625_{10} (multiply by 2)

Multiply	Result	Integer bit
0.625 × 2	1.25	1
0.25 × 2	0.50	0
0.5 × 2	1.00	1

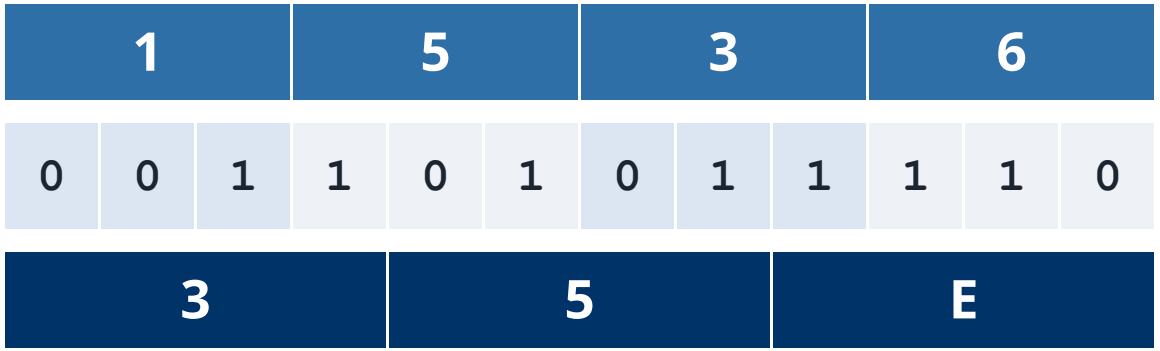
Read bits top → bottom: $.101_2$

$45.625_{10} = 101101.101_2$

Watch out: not every fraction terminates. $0.1_{10} = 0.0001100110011\ldots_2$ repeats forever, which is why computers round decimal fractions.

Octal and Hexadecimal Shortcuts

Octal: 3 bits per digit



Hexadecimal: 4 bits per digit

$$1101011110_2 = 1536_8 = 35E_{16} = 862_{10}$$

Why hex?

- 1 hex digit = 4 bits, so 1 byte = 2 hex digits
- Compact memory addresses, e.g. 0x7FFE1A2C
- Colours: #003468 = R 00, G 34, B 68
- Machine code and MAC addresses

Binary → octal or hex

Group from the binary point outward; pad with zeros

Octal or hex → binary

Replace each digit by its 3-bit or 4-bit pattern

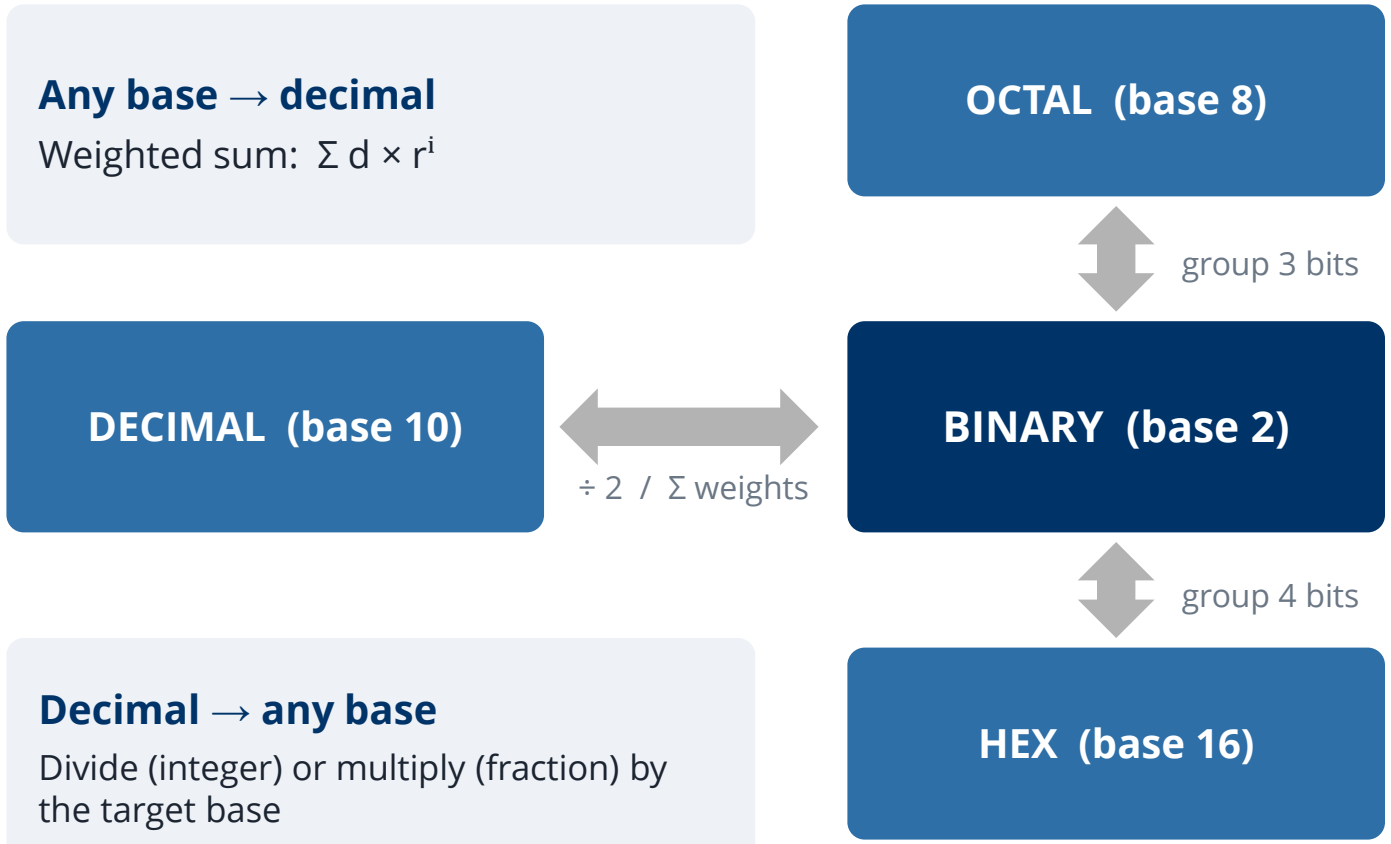
Choosing a Conversion Method

Any base → decimal

Weighted sum: $\Sigma\, d \times r^i$

Decimal → any base

Divide (integer) or multiply (fraction) by the target base



Rule of thumb

- Binary is the hub
- Octal ↔ hex: pass through binary
- Weights work from every base into decimal
- Divide by the target base to leave decimal

Practice Set: Convert by Hand

#	Convert	Method hint	Your answer
1	$110101_2 \rightarrow$ decimal	Weighted sum	
2	$75_{10} \rightarrow$ binary	Repeated ÷ 2	
3	$10111110_2 \rightarrow$ hexadecimal	Group by 4 bits	
4	$2F_{16} \rightarrow$ binary	Hex digit → 4 bits	
5	$745_8 \rightarrow$ decimal	Weights 64, 8, 1	
6	$0.75_{10} \rightarrow$ binary	Repeated × 2	

Challenge: write 255_{10} in binary, octal and hex

Where You Meet These in Real Life

1

IPv4 addresses

192.168.1.10 is four bytes in decimal; 192 = 11000000

2

Web colours

#003468 = red 00, green 34, blue 68, one byte each

3

Memory addresses

Shown in hex, e.g. 0x7FFE1A2C, because 1 hex digit = 4 bits

4

Linux permissions

chmod 755 → 111 101 101 → rwx r-x r-x

5

MAC addresses

00:1A:2B:3C:4D:5E is six bytes written as hex pairs

6

Data dumps

48 65 6C 6C 6F are the ASCII bytes of “Hello”

Common Mistakes to Avoid

✗ Reading remainders from the top when converting decimal to binary	✓ Read remainders bottom to top; the last remainder is the MSB.
✗ Believing n bits can hold the value 2^n	✓ Largest unsigned value is $2^n - 1$ (8 bits → 255).
✗ Forgetting that hex letters A to F stand for 10 to 15	✓ Write the table: A=10, B=11, C=12, D=13, E=14, F=15.
✗ Grouping bits from the left and leaving a short group	✓ Group outward from the binary point and pad: $1011010_2 \rightarrow 0101\ 1010 = 5A_{16}$.

Summary and Next Lecture

- A computer system is hardware and software working through layers of abstraction
- Architecture = instruction set + organization + hardware
- Digital systems use binary because two-state signals are reliable
- Positional value = Σ digit × $radix^{i}$
- Convert with weighted sums, repeated division or multiplication, or bit grouping

Questions?

Next lecture (Hour 2)

Binary arithmetic, complements, signed numbers, data representation, Boolean algebra

Lab

Linux setup, conversion exercises, logic gate simulation

Reading

Mano Ch. 1 (§1-1 to 1-4); Hennessy & Patterson Ch. 1 (§1.1 to 1.3)